Cheng Chen

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Education

Tongji University, Shanghai, China Sept. 2023 – Present
B.Eng. in Electrical Engineering, College of Electronic and Information Engineering
Major Rank: 3/87 | CET-6: 519

Publications

- Yi Xu, **Cheng Chen**, Wenzhuo Lei.
  If You Hear It, Help Find It: Orthogonal Knowledge Distillation for Open-Vocabulary Audio-Visual Event Localization.
  ACM Multimedia (ACM MM) 2026 (Accepted; Oral).
- **Cheng Chen**, Yi Xu, Mufan Cao.
  OrthKD: Extracting Generalized Clinical Knowledge from Heterogeneous Teachers for Lightweight Deployment.
  International Joint Conference on Artificial Intelligence (IJCAI) 2026 (Accepted; Oral).
- Wenzhuo Lei, Mufan Cao, Chaorong Ye, Yi Xu, **Cheng Chen**.
  Enabling Visual Reasoning in Text-Only LLMs via Symbolic Scene Descriptions.
  International Joint Conference on Neural Networks (IJCNN) 2026 (Accepted).
- Yuyao Cai, Yi Xu, **Cheng Chen**.
  Beyond Relevance: Counter-Evidence Planning in a Controlled Budgeted RAG Testbed.
  International Conference on Artificial Neural Networks (ICANN) 2026 (Accepted).
- **Cheng Chen**, et al.
  Neural-Operator Risk Assessment for Deployable Risk Triage.
  Neural Information Processing Systems (NeurIPS) 2026 (Under Review).

Research Experiences

If You Hear It, Help Find It: Orthogonal Knowledge Distillation for Open-Vocabulary Audio-Visual Event Localization
Accepted at *ACM MM 2026*

- Addressed the difficulty of localizing audio-visual events under open-vocabulary queries, where heterogeneous audio and visual teachers provide complementary but uneven supervision and naive distillation propagates modality-specific errors.
- Proposed OvOrthKD, extending orthogonal feature distillation to audio-visual learning so complementary cross-modal cues are preserved while redundant or unreliable teacher signals are suppressed.
- Built an open-vocabulary audio-visual event localization pipeline that remains lightweight and deployable, accepted as an oral presentation at ACM Multimedia 2026.

OrthKD: Extracting Generalized Clinical Knowledge from Heterogeneous Teachers for Lightweight Deployment
Accepted at *IJCAI 2026*

- Addressed expertise asymmetry in multi-teacher distillation for diabetic retinopathy screening, where a weaker ViT teacher (0.83 QWK) harms students when fused through standard logit distillation with a stronger CNN teacher (0.88 QWK).
- Proposed OrthKD, distilling only complementary feature-level context from the weak teacher while enforcing orthogonality between teacher projections to avoid redundant or biased knowledge transfer.
- Evaluated on 132,049 retinal images; the 5.4M-parameter MobileNetV3 student reached 0.885 QWK on EyePACS and improved zero-shot Messidor-2 performance from 0.507 to 0.728 QWK (+22.1 percentage points), enabling edge-deployable screening.

Enabling Visual Reasoning in Text-Only LLMs via Symbolic Scene Descriptions
Accepted at *IJCNN 2026*

- Investigated whether text-only LLMs can perform visual reasoning without multimodal pre-training, avoiding the cost and opacity of end-to-end vision-language models.
- Designed DeepYOLO, which converts images into grammar-constrained symbolic scene reports with an off-the-shelf detector and feeds them to a text-only LLM for question answering.
- On a stratified 5k VQAv2 subset, achieved 42.3% accuracy, outperforming a scene-graph serialization baseline (38.2%) with 2.5× fewer prompt tokens; oracle symbolic inputs reached 44.5%, showing detector quality is the current bottleneck.

Beyond Relevance: Counter-Evidence Planning in a Controlled Budgeted RAG Testbed
Accepted at *ICANN 2026*

- Tackled the limitation of relevance-only retrieval in RAG, where fixed budgets force systems to ignore contradictory evidence and over-trust top-ranked passages.
- Proposed counter-evidence planning in a controlled budgeted RAG testbed that explicitly allocates retrieval budget to both supporting and opposing documents before generation.
- Demonstrated more reliable reasoning under strict retrieval budgets by balancing relevance with explicit counter-evidence search rather than greedy top-k retrieval alone.

Neural-Operator Risk Assessment for Deployable Risk Triage
Submitted to *NeurIPS 2026*

- Addressed the deployment risk of neural operators in scientific computing, where incorrect PDE/operator predictions can silently corrupt downstream analysis without affordable post-hoc verification.
- Developed a single forward-pass risk assessment module that flags unreliable neural-operator outputs at inference time for deployable triage.
- Targeted low-overhead safety screening that can be integrated into neural-operator pipelines without expensive ensemble or sampling-based uncertainty estimation.

Awards

- National 2nd Prize, Global Campus AI Algorithm Elite Competition (2025)
- National 3rd Prize, Embedded Chip & System Design Competition, FPGA track (2025)

Skills

Programming: Python, PyTorch, Git, Linux, LaTeX, Hugging Face, experiment tracking, LLM/agent prototyping.
Mathematics: Linear Algebra, Probability, Optimization, Signal Processing, Mathematical Modeling.
Language: English — CET-6: 519.